

ServiTech, Inc. Agronomy Service Manager (ASM)

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Software Designer

7:06 PM Thu Jul 24 41%

servi tech Agronomy Service Manager [Create PO](#) [Insights](#) [Customers](#) [Logout](#)

Matthew Wildman Agronomist Territory: 41 [Help](#) [FY 2026](#)

Create New Purchase Order

Fill out the form below to create a new purchase order.

Producer and Customer Assignment [Mine](#) [All](#)

Producer*

Select Producer...

[+ Add Customer](#)

Total: 0.0% Must equal 100%

Agronomist Assignment [+ Add Agronomist](#)

Agronomist 1 *	Responsibility %	Amount
Matthew Wildman	100.00	\$ 0.00

Project Overview



Project Description :

Agronomy Service Manager (ASM) is a mobile-friendly, cross-platform web app for ServiTech agronomists to write purchase orders to perform in-field and other services for farmers and cooperatives.



Purpose:

Agronomy Service Manager was developed to replace a legacy Windows application (ACES) used to create purchase orders to perform work for farmers and coops. ACES was over 20 years old and beyond end-of-life.



Role & Responsibilities:

Software Designer for ServiTech, Inc.

- Interviewed key stakeholders and determined feature set
- Defined the database schema
- Designed and coded entire app using Replit's agent.
- Mapped internal database to external API for billing with legacy accounting system



Constraints:

Project duration: June–August 2025

The mandate was to move as quickly as possible and ship a Minimum Viable Product (MVP) before the beginning of the new fiscal year on 10/1/2025

Users and Challenges



Primary Users :

- ServiTech staff agronomists
- ServiTech territory leaders
- Accounts receivable staff



Goals:

- The stated goal of upper management was to quickly create a web app to write purchase orders using any available AI-assisted coding platform to see if it was possible.



Challenges:

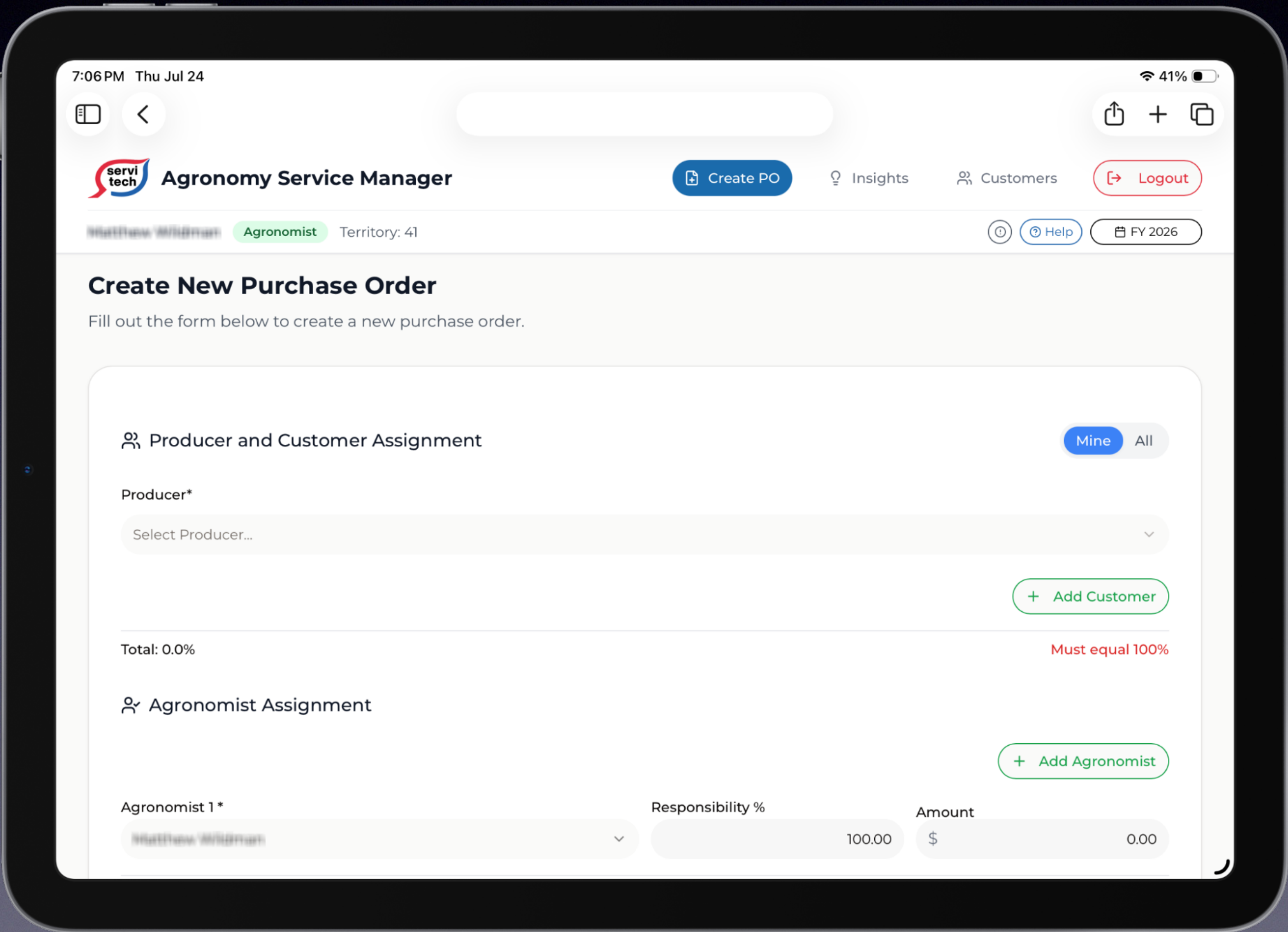
- The biggest challenge was to develop a modern web app and database that could still post data to a legacy database and accounting system.
- It took as long to write and map to that API as it did to write the application itself

Process and Approach

This project differed from previous design projects in terms of the urgency and very short deadline I was given.

Additionally, as instructed, I worked alone in a kind of skunk works mode where I met with key stakeholders, but did not engage with the development team.

From working with agronomists and farmers for nearly 10 years at ServiTech, I was familiar with the existing software and workflows. This made it possible to start designing and coding quickly.



Final Create New PO View



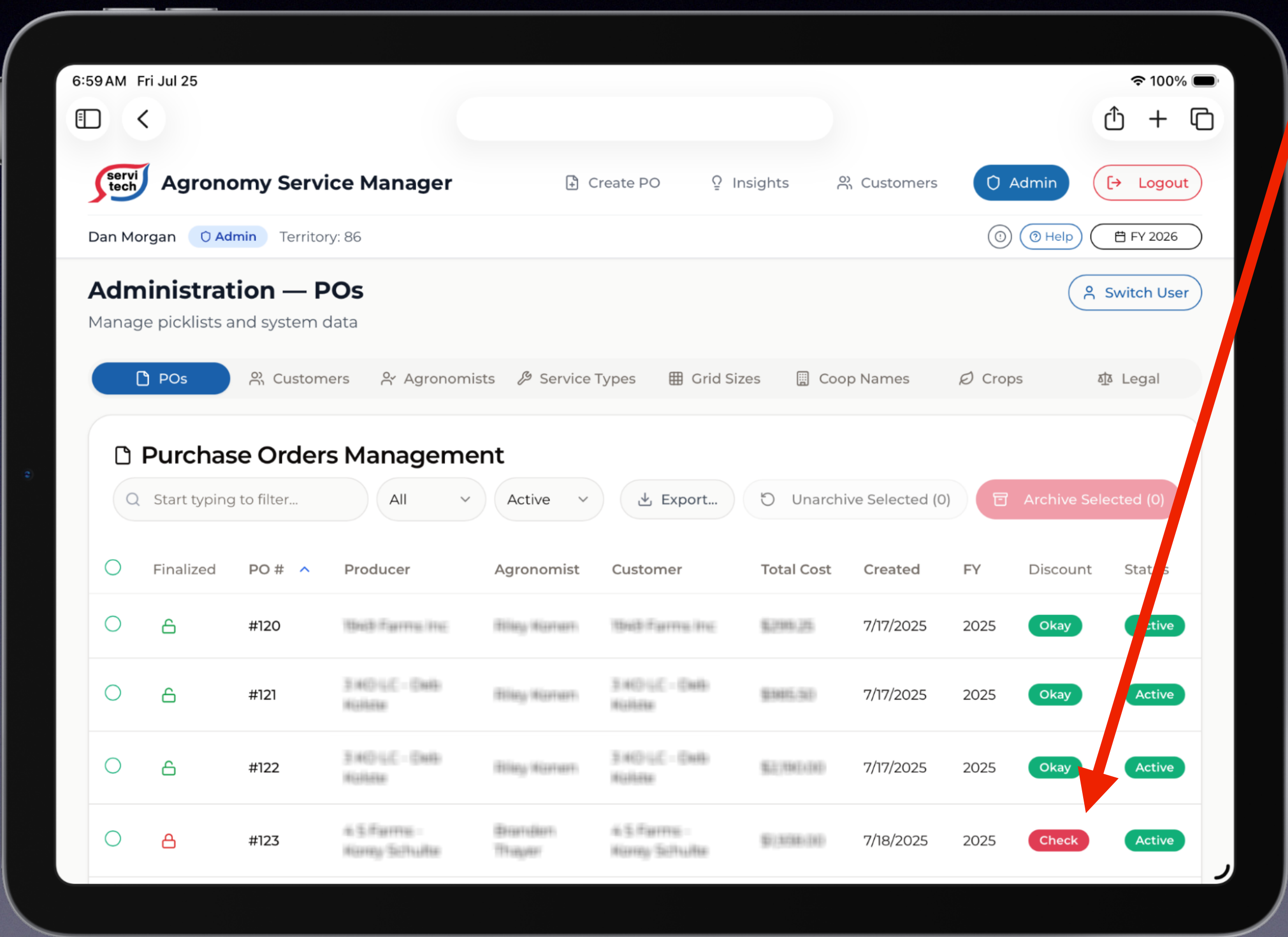
Process and Approach

I started with an initial prompt for the AI agent to begin building the “bones” of the application. I provided the custom CSS classes, instructed it to use Tailwind framework, React, and PostgreSQL for the database.

After the first week of work I met with key stakeholders and gave them the first playback of a working web app.

They provided feedback and a wishlist of features to add and revise. One in particular was an “admin level” permission tier to review all POs that were outside a set threshold for a discount.

PO in violation of discount policy



Final Admin—Review POs View

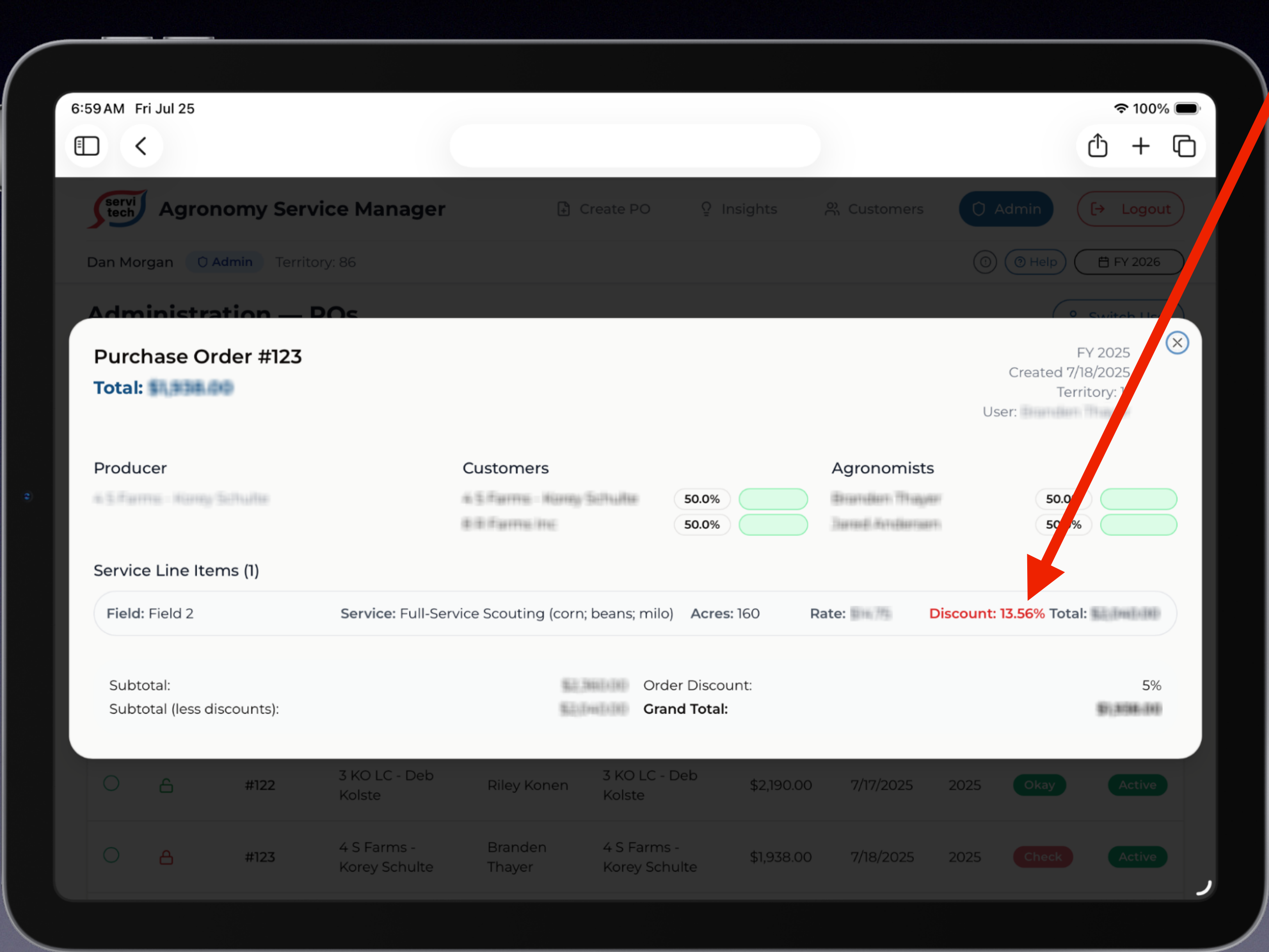


Process and Approach

Any admin or manager permission tier can review any PO.

Drilling down into the details view of PO shows the producer, agronomist splits, billing splits, services line items, and policy violations (in red).

PO in violation of discount policy



Final Admin—Review POs View

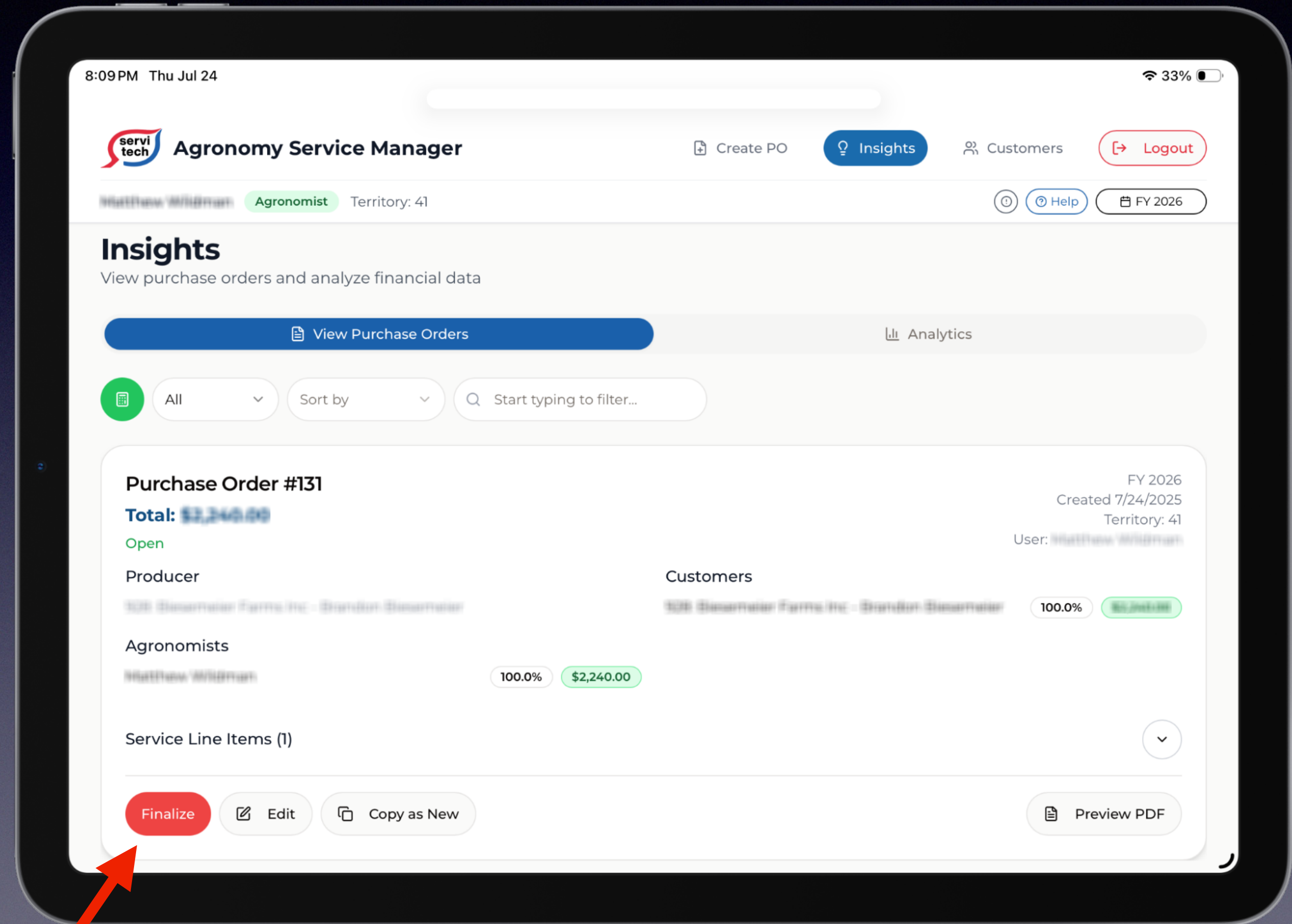


Process and Approach

POs can be edited and saved, but once “finalized” for invoicing, they are locked and immutable.

Additional services are added using a supplemental PO. Changes, such as removal of services, are done through a credit memo.

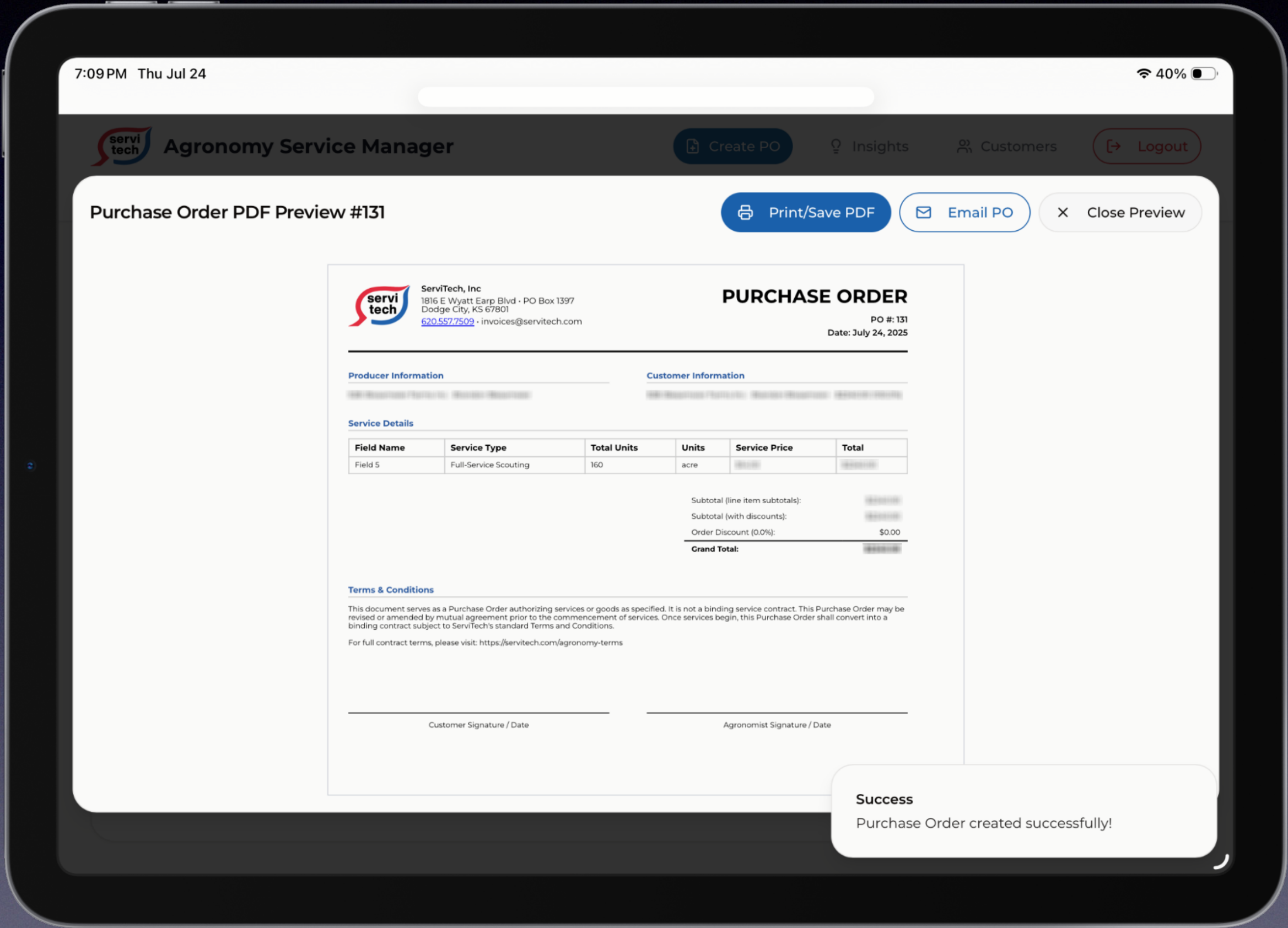
POs are posted to the accounting system through an API to the legacy database.



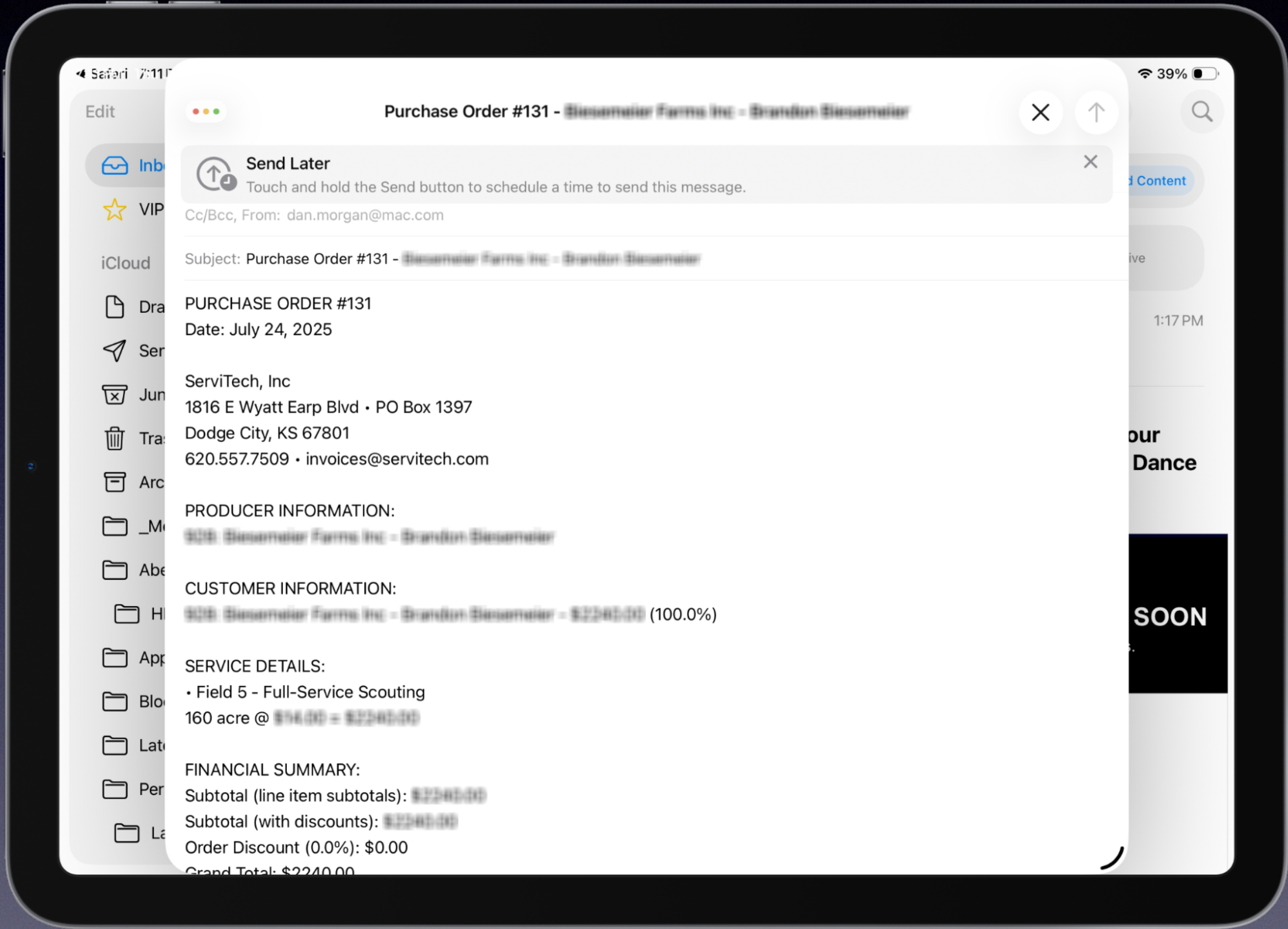
Insights: Review PO and Finalize

Finalize button

Final Designs



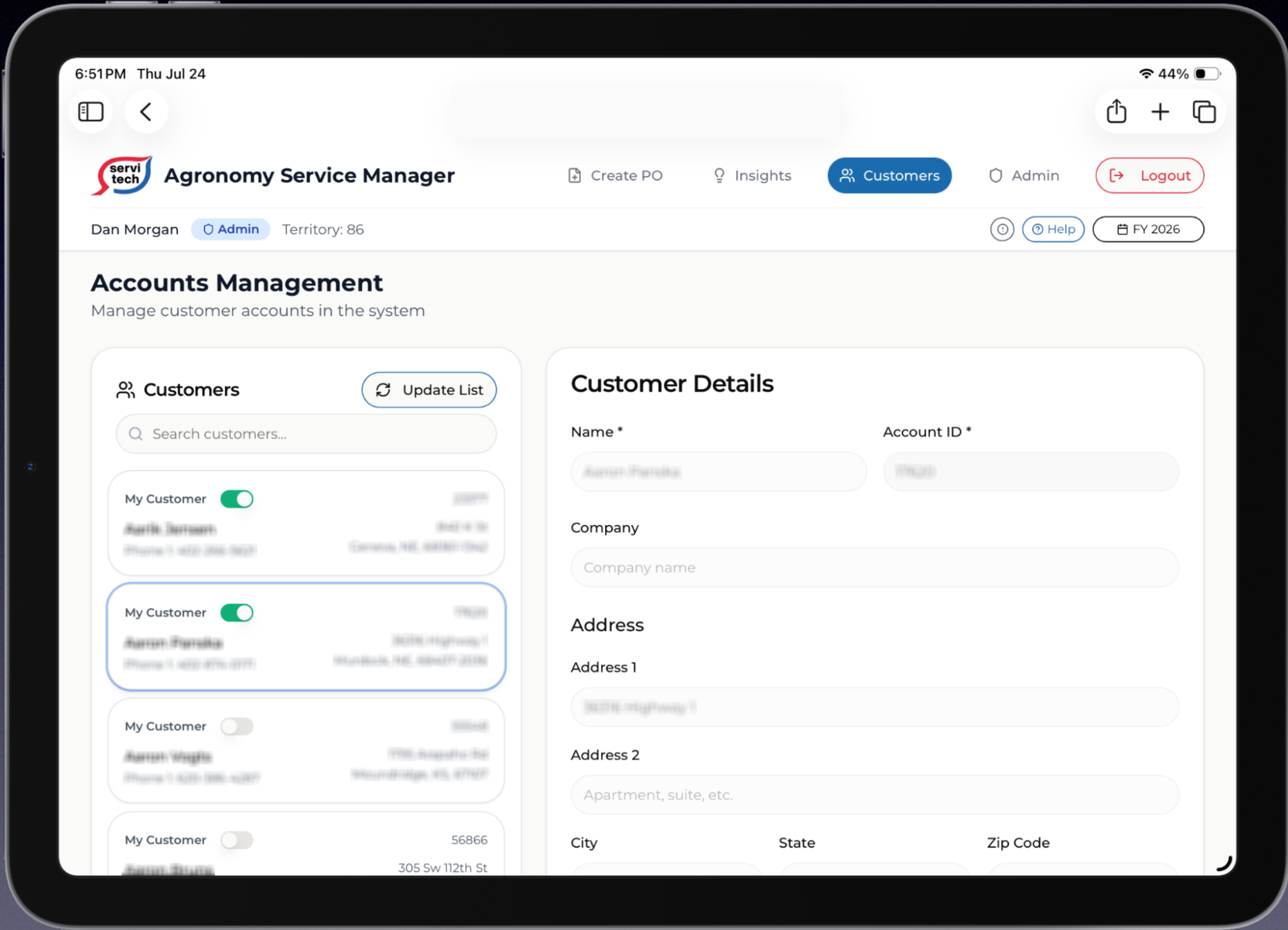
PDF of PO for printing or emailing as attachment



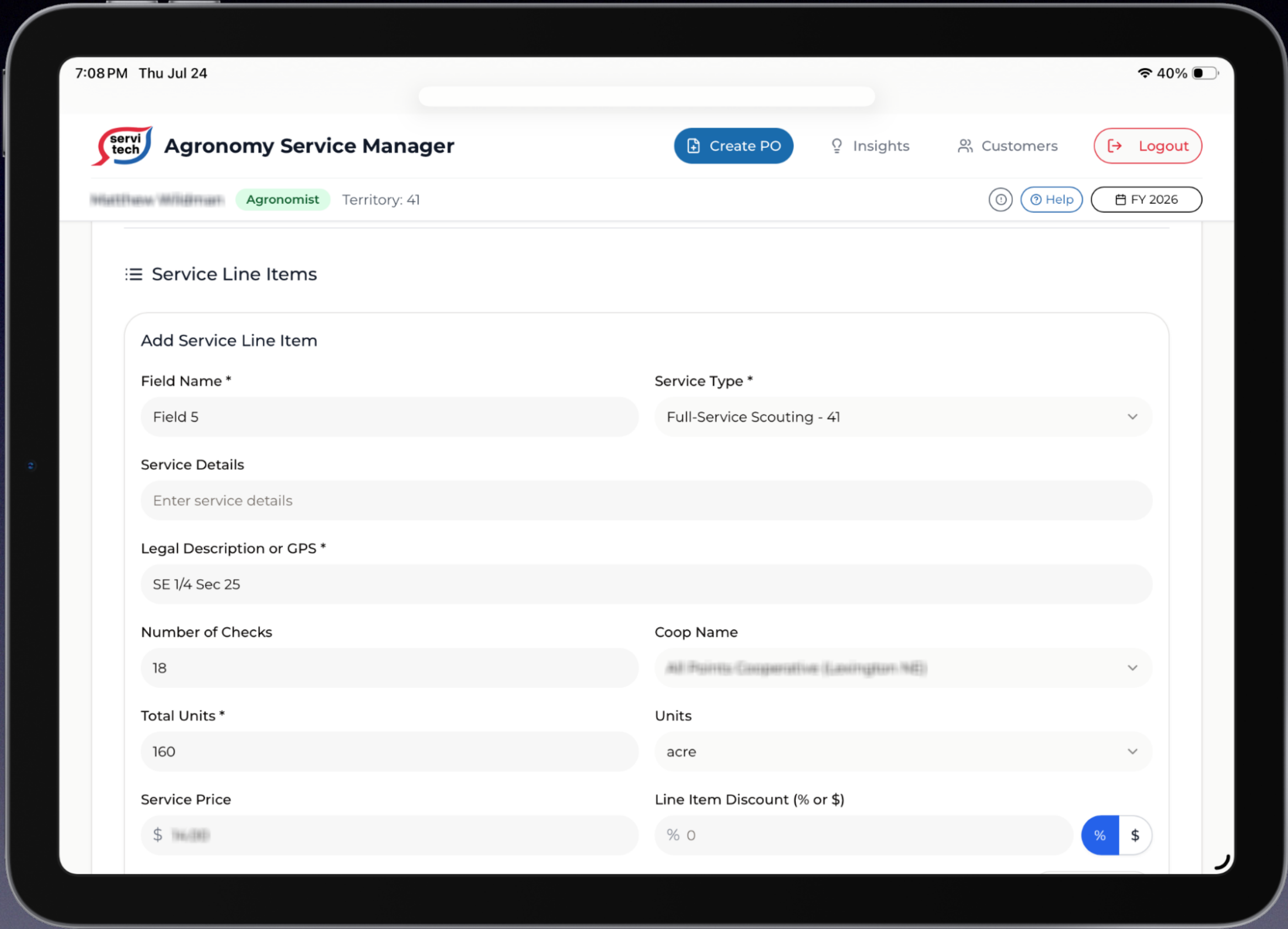
Text version of PO for email



Final Designs



Adding a customer to a User's "My List"



Adding a service line item details

Results and Outcomes

The Agronomy Service Manager (ASM) was available for use in under 60 days from inception in an MVP state.

The Analytics features were pushed off for a future version.

Mapping data from ASM to the legacy accounting system was challenging. I had to put the schemas side by side and map what was required for the accounting system to successfully create an invoice.

Once the connections were determined in a spreadsheet, mapping them using an API created by an IT staffer was straightforward and billing finalize POs was enabled.

